THE INJECTIVE HULL OF A DOMAIN MODULO A PRIME IS INDECOMPOSABLE AND LOCALIZES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The essentialness of the extension of the residue class domain into its fraction field. Invoked inside the counted proof (source lines 826-828 and 839-840). Identifies the injective hull of the quotient with that of the residue field and detects nonzero intersection with submodules.
- The uniqueness of injective hulls and the criterion for an inclusion of injective hulls to be an isomorphism. Invoked inside the counted proof (source lines 828-829, 831-832, 837-838 and 840-841). Identifies hulls computed in different ways and upgrades inclusions of injective submodules to isomorphisms.
- The injectivity of a module over a localization acted on invertibly. Invoked inside the counted proof (source line 834). Shows the hull is injective as a module over the local ring at the prime.

Source reference. Original lemma lemma-essential-extension: sources/stacks-dualizing/pinned-original-source.tex, source lines 371–371. Navigation pointer only: the counted text refers to the Stacks item lemma-essential-extension, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-injective-epimorphism: sources/stacks-dualizing/pinned-original-source.tex, source lines 422–422. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-epimorphism, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-injective-hull-unique: sources/stacks-dualizing/pinned-original-source.tex, source lines 719–719. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-hull-unique, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathfrak{p} \subset R$ be a prime of a ring R. Let E be the injective hull of $R/\mathfrak{p}$. Then*

(1) *E is indecomposable,*
(2) *E is the injective hull of $\kappa(\mathfrak{p})$,*
(3) *E is the injective hull of $\kappa(\mathfrak{p})$ over the ring $R_\mathfrak{p}$.*

Proof. By Lemma lemma-essential-extension the inclusion $R/\mathfrak{p} \subset \kappa(\mathfrak{p})$ is an essential extension. Then Lemma lemma-injective-hull-unique shows (2) holds. For $f \in R$, $f \notin \mathfrak{p}$ the map $f : \kappa(\mathfrak{p}) \to \kappa(\mathfrak{p})$ is an isomorphism hence the map $f : E \to E$ is an isomorphism, see Lemma lemma-injective-hull-unique. Thus E is an $R_\mathfrak{p}$-module. It is injective as an $R_\mathfrak{p}$-module by Lemma lemma-injective-epimorphism. Finally, let $E' \subset E$ be a nonzero injective R-submodule. Then $J = (R/\mathfrak{p}) \cap E'$ is nonzero. After shrinking E' we may assume that E' is the injective hull of J (see Lemma lemma-injective-hull-unique for example). Observe that $R/\mathfrak{p}$ is an essential extension of J for example by Lemma lemma-essential-extension. Hence $E' \to E$ is an isomorphism by Lemma lemma-injective-hull-unique part (3). Hence E is indecomposable. □